

Smart Water Tank: Real-Time Monitoring and Remote Pump Control

Mayank Singh, Electrical Engineering, E-mail: 24110200@iitgn.ac.in
and Mayank Choudhary, Artificial Intelligence, E-mail: 24110199@iitgn.ac.in
Indian Institute of Technology Gandhinagar, Gandhinagar-382355, India

Abstract—This paper presents the design and implementation of a Smart Water Tank Detector that addresses a common household issue: water tank overflow and dry running due to lack of monitoring. The system uses an ESP32 microcontroller, ultrasonic sensor, and flow sensor to monitor water level and flow rate in real time. Additionally, it enables users to remotely control the water pump using the Blynk IoT-based mobile interface, reducing manual errors and water wastage.

I. INTRODUCTION

Urban households often face problems in managing water tanks, including unintentional overflows and pump dry running. This project aims to offer a smart solution by enabling real-time monitoring and control of the tank using IoT technologies. The system integrates ultrasonic level measurement, flow sensing, and remote pump operation through a mobile application.

II. System Design and Methodology

A. Hardware Components

- ESP32 Microcontroller: Main processing and Wi-Fi unit, serves as the brain of the system, providing both processing power and Wi-Fi capabilities. It reads sensor data, performs calculations, and communicates with the Blynk app using virtual pins.
- Ultrasonic Sensor (HC-SR04): Measures water level
- Relay Module: Controls the water pump
- Water Pump: Pumps water into the tank
- Power Adapter: Supplies required voltage
- Jumper Wires and Breadboard: For prototyping and connections.

B. Blynk Dashboard and Virtual Pins

- V1: Water level percentage (Gauge)
- V6: Flow rate in L/min (Value display)
- V5: Numeric input (Target fill %)
- V7: Elapsed time for filling (Display)
- V5: Switch control for relay module (ON/OFF)

The system comprises three main modules: sensing, processing, and control. The sensing module includes an HC-SR04 ultrasonic sensor mounted at the top of the tank to measure distance to the water surface. The ESP32

microcontroller collects data from the sensor, processes it to calculate the percentage fill and estimated time to full capacity, and communicates with the Blynk platform.

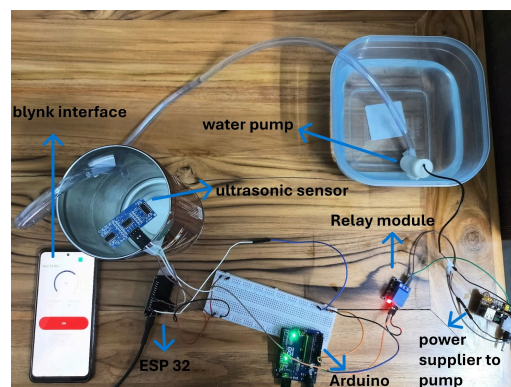


Fig 1. Components involved in the Project

A relay module is used to control the water pump, which the user can switch ON/OFF remotely using a button in the mobile interface.

III. IMPLEMENTATION

The hardware includes the ESP32, ultrasonic sensor, and a relay-controlled pump. Software was developed in Arduino IDE using the Blynk, WiFi, and sensor libraries. Key virtual pins in the Blynk dashboard include: V1 for water level percentage, V6 for flow rate, V7 for estimated fill time. Sensor readings are collected and sent to Blynk every second using BlynkTimer. A dashboard in Blynk Cloud was created with gauges, display, and a switch widget to visualize and interact with these parameters. The project uses Arduino IDE for programming the ESP32. Sensor pins are connected to designated GPIOs:

- Ultrasonic Sensor: TRIG → GPIO12, ECHO → GPIO14
- Relay Control: GPIO5

Using Blynk's virtual pins, data is updated every second. The relay is toggled ON/OFF using a switch button. Real-time data for flow rate, percentage filled, and estimated time is visualized on the Blynk dashboard.

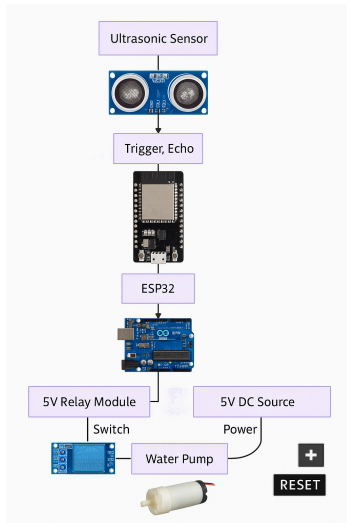


Fig. 2 Circuit diagram of our project

These are the major components of our project:

A. ESP 32 Microcontroller

In this Smart Water Tank Detector system, the ESP32 acts as the **central control unit**. It was chosen over Arduino Uno because:

- It supports Wi-Fi, which is essential for real-time communication with the Blynk app.
- It has enough processing power to handle multiple sensor inputs and live calculations (water level, flow rate, elapsed time).
- Its GPIO availability allows connections for the ultrasonic sensor, flow sensor, and relay simultaneously.
- It's cost-effective and compact, making it ideal for home automation.

Receives data from the ultrasonic sensor (via TRIG and ECHO pins) to calculate tank water level. Reads pulses from the YF-S201 flow sensor to determine flow rate. Controls the relay module through GPIO to switch the pump ON/OFF. Connects to Wi-Fi and communicates with the Blynk IoT platform. Processes logic to compute the percentage filled and estimate the time to full, and sends this data to the phone app using virtual pins.

B. Ultrasonic sensor

The ultrasonic sensor is mounted at the top of the water tank. It sends a short pulse from the TRIG pin and listens for the return echo via the ECHO pin. The ESP32 calculates the time difference, converts it to distance, and subtracts that from the total tank height to find the current water height. From this height, the code calculates the percentage of the tank filled. This data is sent to the Blynk app to show a real-time gauge of water level on the user's phone.

C. Relay module

A relay module is an electrically operated switch that allows a low-power device like a microcontroller to control high-voltage or high-current loads, such as motors or pumps. In this project, a 5V single-channel relay is used to turn the water pump ON or OFF based on instructions from the ESP32.

- The ESP32 cannot directly power a high-voltage pump, but it can trigger the relay to do so. It allows safe switching between low-power control logic and high-power pump hardware.
- The relay is connected to GPIO5 of the ESP32. When the user taps the switch in the Blynk app, a signal is sent to ESP32. ESP32 sets the GPIO HIGH or LOW to turn the relay ON or OFF. The relay switches the power to the water pump, starting or stopping the water flow based on the fill level set by the user.

IV. EXPERIMENTAL RESULTS

The system was tested in a real-world setup. It accurately measured water level in percentage format and provided reliable flow rate data. The estimated time to full was dynamically updated based on flow. The mobile application allowed seamless control of the water pump. The prototype performed reliably during continuous operation and demonstrated the potential for residential application. The prototype system successfully displayed real-time tank fill percentage, flow rate, and elapsed filling time. The relay could control the water pump from the phone app as per user input.

V. Conclusions

The Smart Water Tank Detector proves to be a practical and effective solution to water management issues in households. It automates monitoring and control processes, minimizes water wastage, and offers user-friendly remote access via a smartphone. The project demonstrates successful integration of sensors, microcontroller processing, and IoT communication in a real-time application. It provides a practical and efficient method for managing household water tanks. It simplifies monitoring, saves water, and allows remote pump control via a smartphone. Future versions may include water quality detection or voice assistant integration.

VI. ACKNOWLEDGEMENT

We would like to thank the faculty and lab staff of ES116 for their continuous guidance and support.

VII. REFERENCES

- [1] [Blynk IoT Platform Documentation](#):
- [2] [HC-SR04 Ultrasonic Sensor Datasheet](#)